SHUAI ZHOU

shuaizho@cs.cmu.edu | shuai-zhou.com | ShuaiZhou302

Educations

Carnegie Mellon University — **Pittsburgh, USA**
Master of Science in Robotics — *Aug 2026 — Jun 2028*
Advisor: ***Zackory Erickson***

South China University of Technology — **Guangzhou, China**
Bachelor of Engineering in Robotics — *Aug 2022 — Jun 2026*

Carnegie Mellon University — **Pittsburgh, USA**
Visiting Student in the Robotics Institute — *Jun 2025 — Mar 2026*
Advisor: ***Jiaoyang Li, Ding Zhao***

University of California, Berkeley — **Berkeley, USA**
Exchange Student in EECS — *Aug 2023 — Dec 2023*

Publications (* equal contribution)

Preprints

[P4] **ActiveScale: Scaling Active Perception for Robots across Model, Data, and Hardware**
Shuai Zhou*, Kaisheng Pang*, Wenxuan Song, Wenjie Zhang, Xinhu Zheng, Haoang Li.
Under Review, 2026 [Paper]

[P3] **UniWAM: Unified World-Action Model**
Jiayi Chen*, Wenxuan Song*, Jingbo Wang*, Shuai Zhou*, Xicheng Gong*, Zehua Fan, Ziyang Zhou, Junwu E, Haodong Yan, Fuhao Li, Qize Yu, Xu Huang, Pengwei Wang, Wen Chen, Shunbo Zhou, Haoang Li.
Under Review, 2026 [Paper]

[P2] **From Discrete Plans to Real-World Execution: A World-Model-Driven Framework for Execution-Aware Multi-Agent Path Finding**
Jingtian Yan, Shuai Zhou, He Jiang, Stephen Smith, Jiaoyang Li.
Under Review, 2026 [Paper]

[P1] **EgoX: Egocentric Cross-Embodiment Manipulation with Embodiment Dreaming**
Yaru Niu*, Binghong Chen*, Changwei Yao*, Shangtao Li*, Zhenlong Fang*, Revanth Krishna Senthilkumaran, Shuai Zhou, Yuemin Mao, Kalie Ching, Po-Ju Chao, Yudong Luo, Peiqi Yu, Heng Zhang, Hao Zhang, Bingqing Chen, Chen Qiu, Hongtei (Eric) Tseng, Changliu Liu, Jonathan Francis, Ding Zhao.
Under Review, 2026 [Paper]

Conference Proceedings

[C4] **LAMP: Long-Horizon Adaptive Manipulation Planning for Multi-Robot Collaboration in Cluttered Space**
Shuai Zhou, Yorai Shaoul, Jiaoyang Li.
IROS, 2026. [Paper]

[C3] **HTD: Learning Versatile Humanoid Manipulation with Touch Dreaming**
Yaru Niu, Zhenlong Fang, Binghong Chen, Shuai Zhou, Revanth Krishna Senthilkumaran, Hao Zhang, Bingqing Chen, Chen Qiu, Hongtei (Eric) Tseng, Jonathan Francis, Ding Zhao.
IROS, 2026. [Paper]

[C2] **LSRP*: Scalable and Anytime Planning for Multi-Agent Path Finding with Asynchronous Actions**
Shuai Zhou, Shizhe Zhao, Zhongqiang Ren.
SoCS, 2025. [Paper]

[C1] **Loosely Synchronized Rule-Based Planning for Multi-Agent Path Finding with Asynchronous Actions**
Shuai Zhou, Shizhe Zhao, Zhongqiang Ren.
AAAI, 2025. [Paper]

Experiences

Carnegie Mellon University — **Pittsburgh, USA**
Graduate Research Assistant, advised by Zackory Erickson — *Aug 2026 — Present*
Research Topics: Dexterous Manipulation, Representation Learning.

- Action tokenisation for generative dexterous manipulation control.
- Dexterous manipulation for tool use with curriculum learning.

Hong Kong University of Science and Technology (Guangzhou) — **Guangzhou, China**
Research Assistant, advised by Haoang Li — *Mar 2026 — Present*
Research Topics: Robot Foundation Model, Active Perception, Egocentric Human Data.

- Scaling Active Perception VLA with AMP teleoperation system and Human-Robot co-training [P4].
- World action model pretraining with egocentric-human data and corresponding annotation pipeline [P3].

CARNEGIE MELLON UNIVERSITY **Pittsburgh, USA**
Visiting Researcher, advised by Ding Zhao *Aug 2025 — Mar 2026*
Research Topics: Humanoid, Egocentric Human Data, Whole-Body Controller.

- Learning versatile humanoid manipulation policy and developing robust lower-body controller [C3].
- Egocentric cross-embodiment manipulation with embodiment-specific latent prediction [P1].

CARNEGIE MELLON UNIVERSITY **Pittsburgh, USA**
Visiting Researcher, advised by Jiaoyang Li *Apr 2025 — Mar 2026*
Research Topics: Learning for Planning, Multi-Robot System.

- Collaborative multi-robot non-prehensile manipulation via flow-matching model and lazy planning [C4].
- Learning execution-time prediction models for closed-loop multi-robot coordination and dependency optimization [P2].

SHANGHAI JIAO TONG UNIVERSITY **Shanghai, China**
Research Intern, advised by Zhongqiang Ren *Apr 2024 — Apr 2025*
Research Topics: Heuristic Search, Multi-Robot System.

- Scalable planners for multi-robot path planning that scale to 1,000+ robots [C1].
- Complete, anytime planning framework with pruning and sorting strategies that accelerated search [C2].

SKILLS

Platforms: Wuji Hand 2, X-Arm, AgileX ALOHA, AgileX Piper Arm, Apple Vision Pro, Meta Quest 2, Unitree G1 Humanoid
Simulators: MJLab, IsaacLab, MuJoCo, PyBullet
Programming: Python, C++, Java, HTML, MATLAB, LaTeX

SERVICES

Conference Reviewer:
2025-2026 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)
2026 The Conference on Robot Learning (CoRL)
2027 IEEE International Conference on Robotics & Automation (ICRA)
Journal Reviewer:
2026 IEEE Robotics and Automation Letters (RA-L)

HONORS

Merit Student of South China University of Technology
Top student in Robotics Engineering, Class of 2022

Best Paper Award
IROS 2026 PWMS(Physical World Models for Scaling Embodied AI) Workshop

PRESS COVERAGE

"LAMP Helps Robots Find a Way Through"
by Mallory Lindahl, CMU Robotics News, 2026 [Link]